

TechCatalog — End-to-End E-Commerce Data Pipeline

An automated Azure-native pipeline turning scattered e-commerce data into business-ready insights.

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1. WHY THIS PROJECT EXISTS

- **The Problem:** Electronics product and price data is scattered across multiple e-commerce APIs (e.g., Amazon, Google Shopping), each with its own unstructured payload and format — leaving it far from analysis-ready for stakeholders.
- **The Objective:** Build an automated end-to-end data pipeline that collects, processes, cleanses, and models this multi-source data, turning it into business-ready insights inside Power BI.

2. SYSTEM ARCHITECTURE & DATA PLATFORM

TechCatalog is built on a modern, Azure-native cloud data platform:

ORCHESTRATION **Orchestration & Monitoring** — Managed via Azure Databricks Workflows running daily scheduled job triggers (10:00 AM Riyadh time).

INGESTION **Ingestion Layer** — Python REST requests via RapidAPI Gateway with token-based authentication, retrieving daily product and price payloads across 5 electronics categories.

STORAGE **Storage & Processing Layer** — Azure Data Lake Storage Gen2 (ADLS Gen2) paired with Databricks Delta Lake, implementing a Medallion Architecture (Bronze → Silver → Gold).

SECURITY **Data Security** — Encrypted API keys and storage credentials managed through Azure Key Vault and Databricks Secret Scopes.

BI **Business Intelligence** — Interactive Power BI Dashboard fed by Gold layer star-schema Delta tables.

3. MEDALLION ARCHITECTURE & DATA PROCESSING

Step 1 — Ingestion (Bronze Layer)

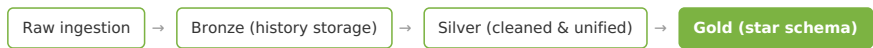
- Fetches raw JSON payloads daily via Python REST calls from Amazon & Google Shopping APIs across 5 electronics categories.
- Stores timestamped raw JSON files directly in **ADLS Gen2 (Bronze Path)** in append mode for historical auditing and tracking.

Step 2 — Transformation & Standardization (Silver Layer)

- **PySpark Processing:** Performs deduplication, price string parsing, data type casting, and schema enforcement.
- **Data Enrichment:** Extracts brand names and handles missing/null values for ratings and availability.

Step 3 — Star Schema Data Modeling (Gold Layer)

- Organizes clean data into optimized Delta tables forming a **Star Schema**:
- **Fact Table:** gold_fact_products (Price, rating, discounts, market metrics).
- **Dimension Tables:** gold_dim_product & gold_dim_logistics.



4. PIPELINE AUTOMATION & ORCHESTRATION

- **Databricks Workflows (DAG Execution):** Sequential task execution enforcing strict dependencies: Bronze Notebook → Silver Notebook → Gold Notebook on serverless Databricks compute.
- **Enterprise Reliability:** Automatic retry mechanisms for API rate limits, failure logging, and automated refreshes for the downstream Power BI reports.

5. BUSINESS INTELLIGENCE & POWER BI ANALYTICS

The Gold Star Schema powers an executive Power BI dashboard containing:

50K+

total products cataloged

2.20K SAR

average product price

4.12 ★

average customer rating



Market & Price Benchmarking Dashboard — Visual Preview

50K+ Total Products

2.20K SAR Avg Price

4.12 ★ Avg Rating

POWER BI DASHBOARD LAYOUT — SCREENSHOT PLACEHOLDER

- **Price & Market Analytics:** Category price distribution, brand price benchmarking (comparing Dell, Asus, Apple, Acer, HP, Lenovo, Samsung, Microsoft), and Price vs. Rating correlation scatter plots.
- **Dynamic Slicers:** Interactive filtering by Market, Product Category (Laptop, Phone, Tablet, TV, iPad, Other), and Brand.

6. DATA GOVERNANCE & DEVOPS

Data Quality

Enforced via Delta Lake ACID transactions — every batch is validated before commits.

Data Layers

Unidirectional flow: Raw → Bronze → Silver → Gold, with no direct manual edits allowed.

Access & Secrets

Governed by Unity Catalog permissions and Azure Key Vault scopes.

Versioning & Ops

Orchestrated single-workflow pipeline with versioning managed on GitHub.

HOW IT COMPARES

CAPABILITY / FEATURE	TECHCATALOG PIPELINE	TRADITIONAL MANUAL ETL	BASIC BI CONNECTORS
Automated End-to-End Pipeline	✓ Yes (Serverless DAG)	✗ No (Manual)	Partial
Medallion Delta Lake Architecture (Bronze → Silver → Gold)	✓ Yes	✗ No	✗ No
Encrypted Secret Management (Azure Key Vault)	✓ Yes	✗ No (Hardcoded)	Limited
Star Schema Analytics Ready	✓ Yes (Gold Fact & Dim Tables)	✗ No	Limited
Automated Fault Tolerant Retry & Logging	✓ Yes	✗ No	✗ No

| 7. RETROSPECTIVE: CHALLENGES, SOLUTIONS & FUTURE ROADMAP

Challenges Faced

- API schema inconsistencies across Amazon & Google Shopping.
- Azure ADLS Gen2 storage access permission issues.
- Databricks session state and execution dependencies.

Solutions Implemented

- Unified PySpark data normalization and schema standardization routines.
- Reconfigured Azure Key Vault secret scopes and access policies.
- Structured sequential notebook execution DAGs in Databricks Workflows.

Future Roadmap

- **Historical Price Trend Analysis:** Track price fluctuations over extended timelines.
- **Predictive Price Analytics:** ML models to predict competitor price drops.
- **Demand Forecasting:** Integrate product demand and search trend signals.

| PROJECT TEAM

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